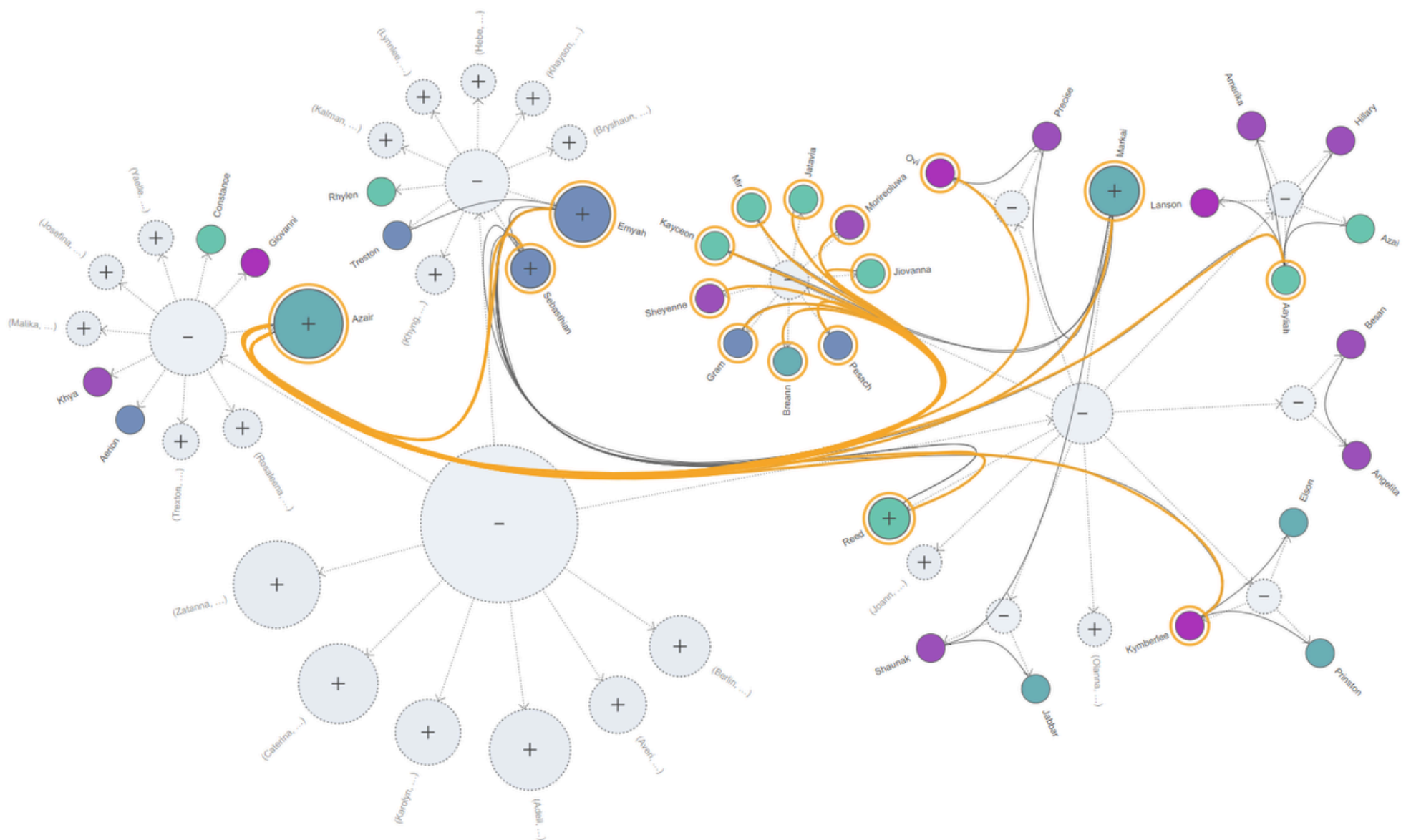


Living Knowledge Graph

A breakdown for engineers **building agent memory**, not just chatbots



Swipe for the full technical breakdown

The Starting Point

A standard **knowledge graph is static**

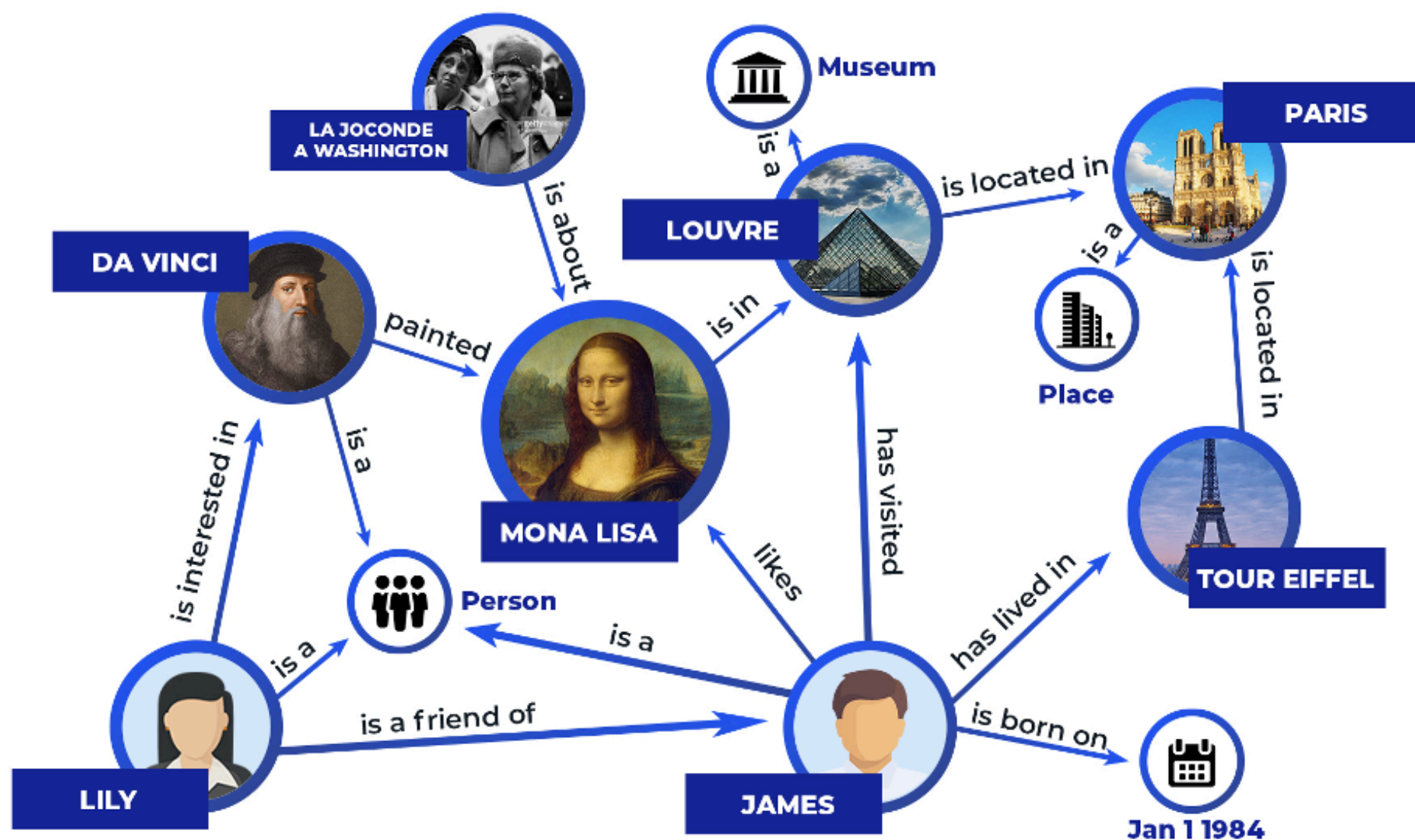


Image Source: zilliz.com

It stores entities and relationships as if they are permanently true. Facts get written once and treated as fixed.

Problem: the real world changes. A fact that was true last month may be false today, and a static graph has no way to represent that shift.

The Definition

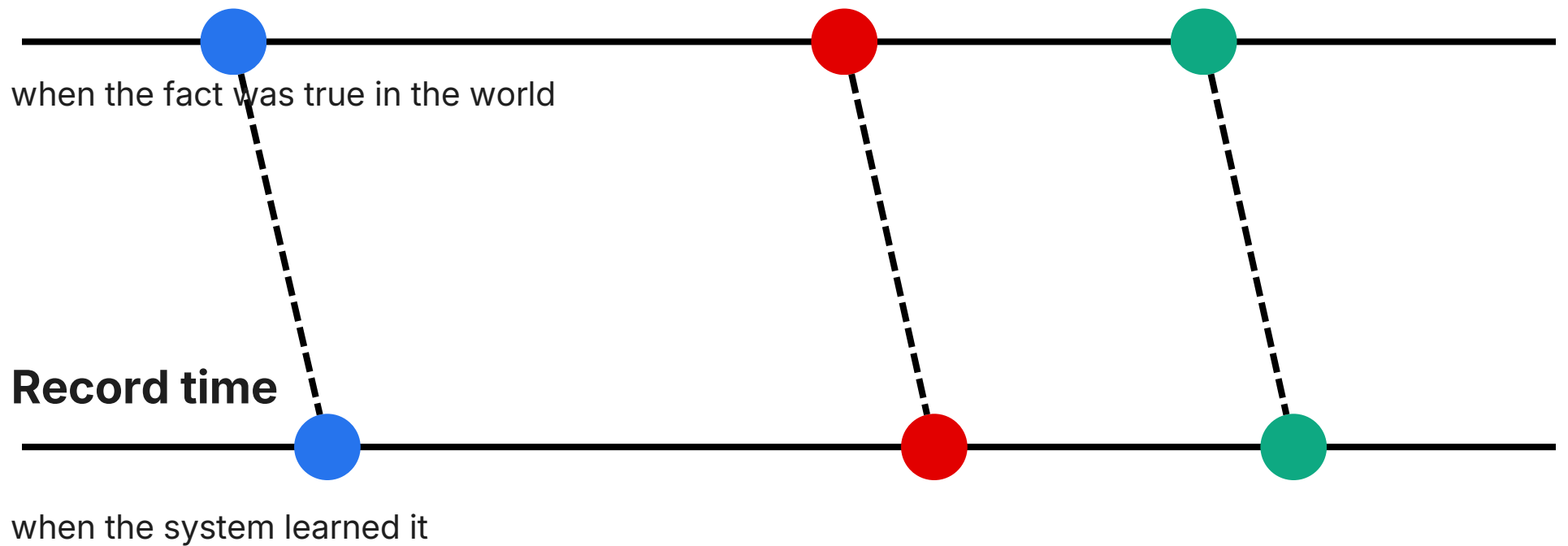
A living knowledge graph updates itself continuously as new information arrives, instead of being rebuilt from scratch.

- Ingests new facts in real time, from conversations, documents or agent output
- Keeps a full history of how knowledge changed, not just the latest state
- Lets agents reason about what is true now versus what used to be true

The Core Mechanism

Bi-temporal tracking: two clocks, not one

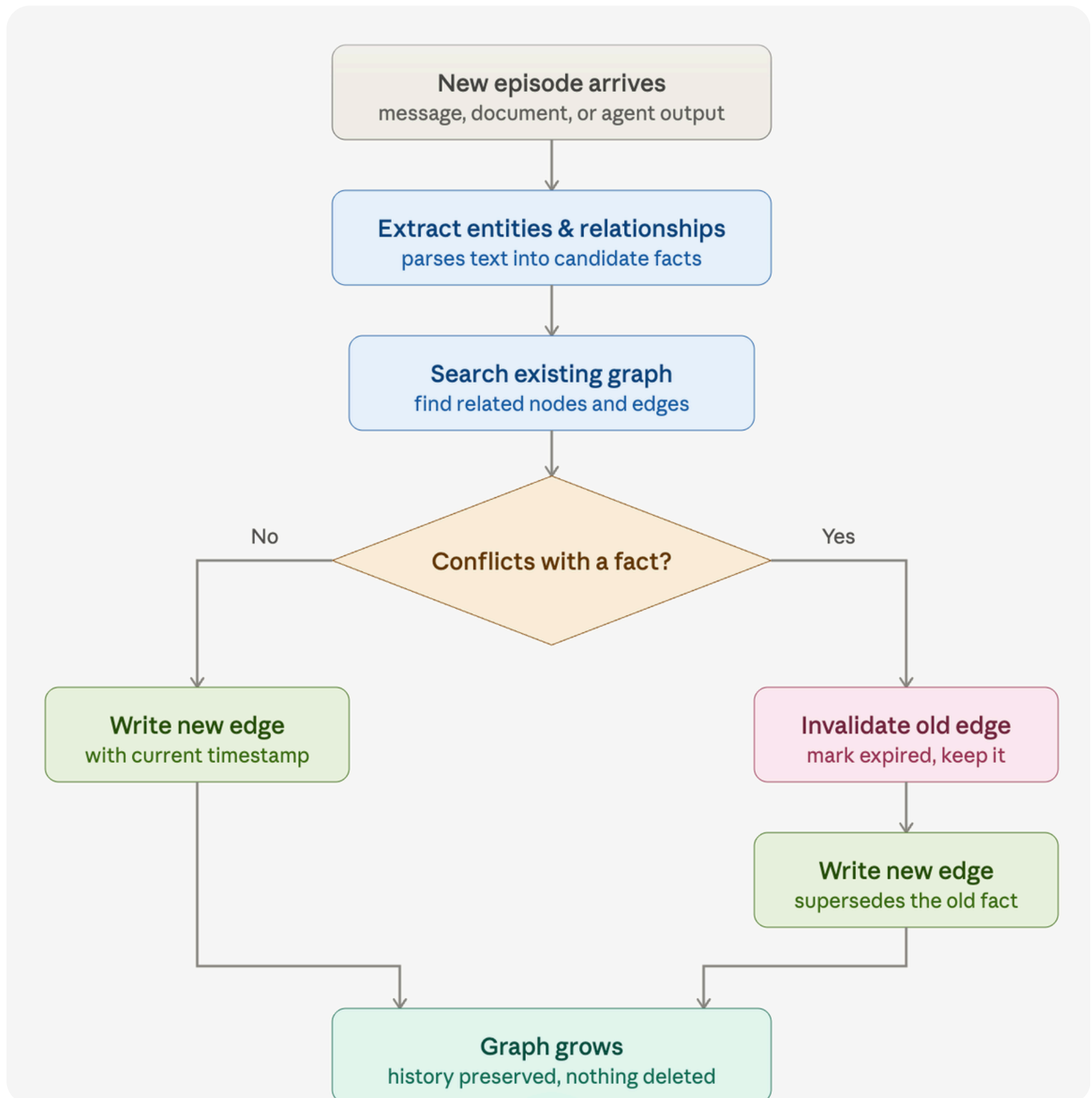
Valid time



Old facts are marked invalid, not deleted. The graph keeps a full audit trail of what changed and when.

How it updates

The read-write loop behind a living graph



Why it matters for Agents

Plain chat history and RAG fall short here

Chat history

Grows linearly, no structure, hard to query for specific facts

Static RAG

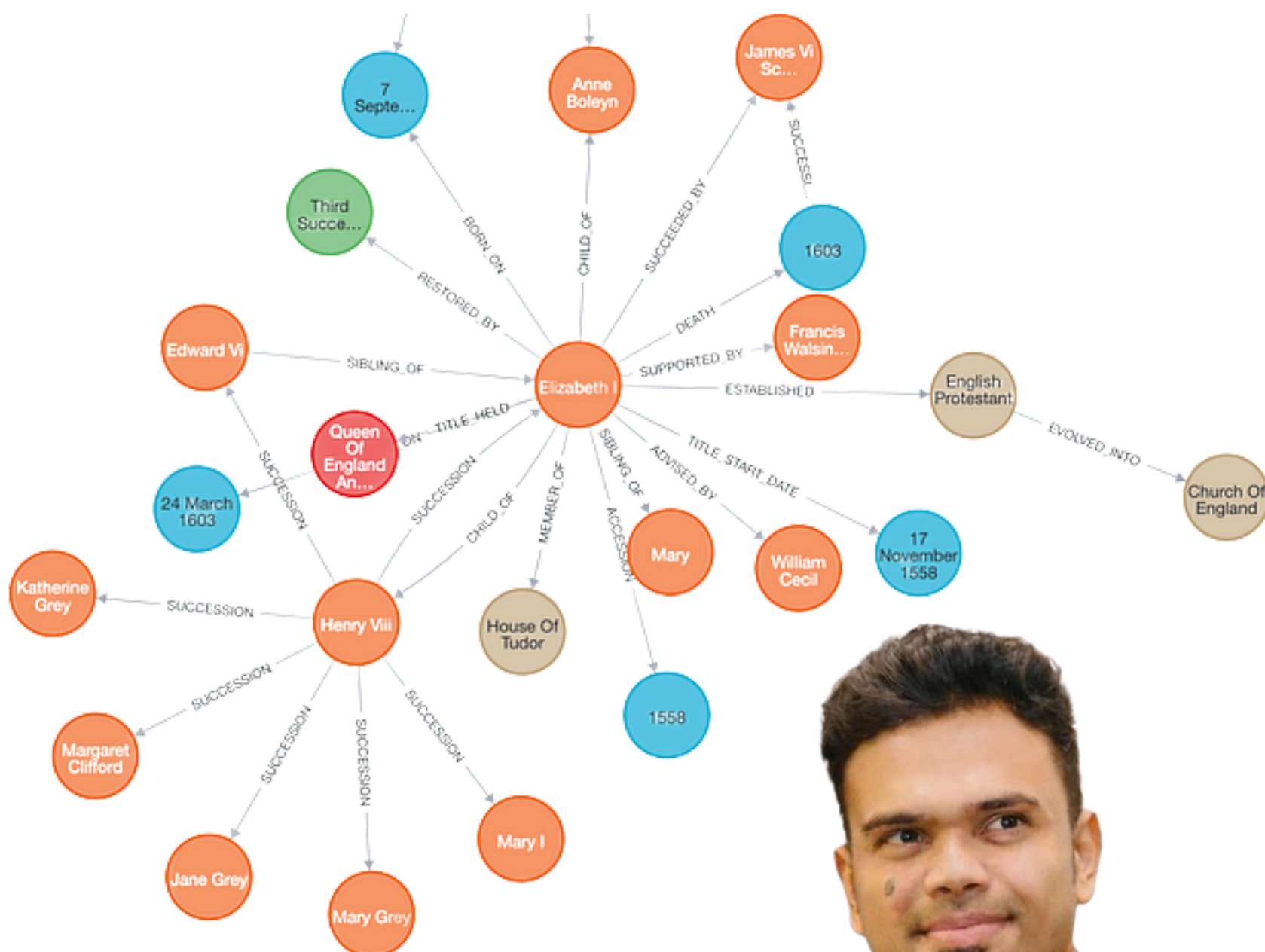
Retrieves similar text chunks, cannot represent change over time

Living knowledge graph

Structured, timestamped, queryable, explainable, and grounded

SEE LIVE AT DHS 2026

This is the backbone of the movie intelligence hack session. In "**When Agents Meet Living Graphs**," a bi-temporal graph captures every idea, approval, rejection and revision a director makes.



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